

Course Code	Name of the Course			
216U03C302	Basic Electronic Circuits			
Teaching Scheme	TH	P	TUT	Total
	03	-	-	03
Credits Assigned	03	-	-	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	-	20	30	50
				100

Course pre-requisites:

- Elements of Electrical and Electronics Engineering

Course Objectives:

The objective of the course is to impart fundamental knowledge and applications of semiconductor devices like P-N junction diode, BJT and FET. This course aims to build a foundation for DC analysis, biasing circuits of BJT, FET and small signal analysis of mid frequency range amplifiers using hybrid π model.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Analyze and design Diode circuits.
- CO2.** Analyze the transistor circuits for DC operation.
- CO3.** Understand working of transistor as an amplifier.
- CO4.** Analyze transistor amplifier for small signal operation for mid frequency range.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Diode Applications		07	CO1
	1.1	Review of p-n junction diode characteristic and current-voltage relationship		
	1.2	Voltage transfer characteristics, series and shunt clipper, diode series and shunt clamper		
	1.3	Working and applications of Varactor diode, Schottky diodes		
2	Bipolar Junction Transistor (BJT)		09	CO2
	2.1	Review of BJT characteristics, regions of operation. DC load line, Early effect and thermal runaway		
	2.2	DC Analysis and design of Fixed bias, voltage divider bias and Collector to base bias, bias stabilization for fixed bias and voltage divider bias, Stability factor $S(\beta)$		
	2.3	BJT configurations: CE, CB, CC		
	2.4	BJT as a switch		
3	Field Effect Transistors (FET)		09	CO2
	3.1	MOSFET: Structure, working, characteristics, two terminal basic MOS capacitor structure, nonideal current voltage characteristics: finite output resistance, Body effect		
	3.2	DC circuit analysis, load line and modes of operation and design of biasing circuits of MOSFET		
	3.3	MOSFET configurations: Common Source		
	3.4	MOSFET as a switch		
4	BJT Amplifiers		12	CO3, CO4
	4.1	Bipolar linear amplifier, small signal Hybrid π equivalent circuit of BJT		
	4.2	Small signal mid-frequency equivalent circuit - hybrid-pi model		
	4.3	Small signal mid-frequency analysis of CE, CB and CC amplifier using hybrid-pi model for voltage gain, input and output impedance		
5	FET Amplifiers		08	CO3, CO4
	5.1	Load lines and small signal parameters, Small signal equivalent circuit		
	5.2	Analysis of Enhancement MOSFET CS, CG, CD amplifier using hybrid-pi model for voltage gain, input and output impedance		
Total			45	

Reference Books

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	D. A. Neamen	<i>Electronic Circuit Analysis and Design</i>	McGraw Hill Education, India	3rd Edition 2014
2	Boylestad and Nashelesky	<i>Electronic Devices and Circuits Theory</i>	Pearson Education, India	10th Edition 2009
3	Thomas L. Floyd	<i>Electronic Devices</i>	Prentice Hall (Pearson)	10th Edition 2016
4	Adel S. Sedra, Kenneth C. Smith, Arun N Chandorkar	<i>Microelectronic Circuits Theory and Applications</i>	OXFORD International Edition, India	6th Edition, 2013